

Project Documentation

# DEPLOYMENT OF NGINX WEB SERVER ON UTHO CLOUD INFRASTRUCTURE

Created By  
**Suryakant Kulkarni**



## ABOUT UTHO CLOUD

Utho is an Indian cloud service provider that offers virtual machines, storage, networking, and security services. It provides scalable and cost-effective cloud infrastructure through a user-friendly dashboard, making it suitable for developers, startups, and businesses.

## PROJECT OVERVIEW

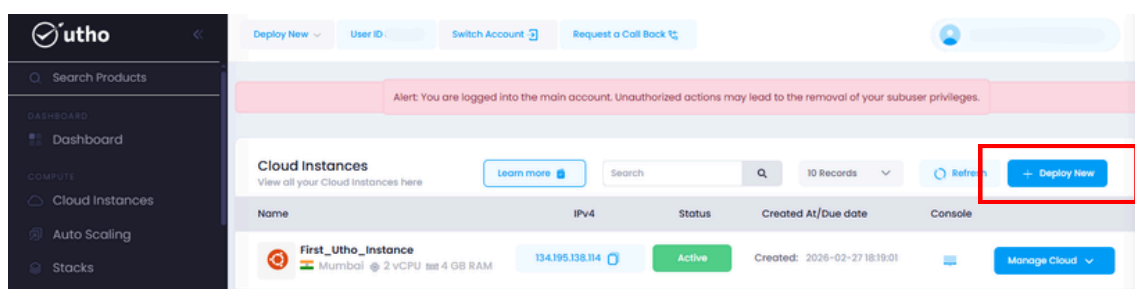
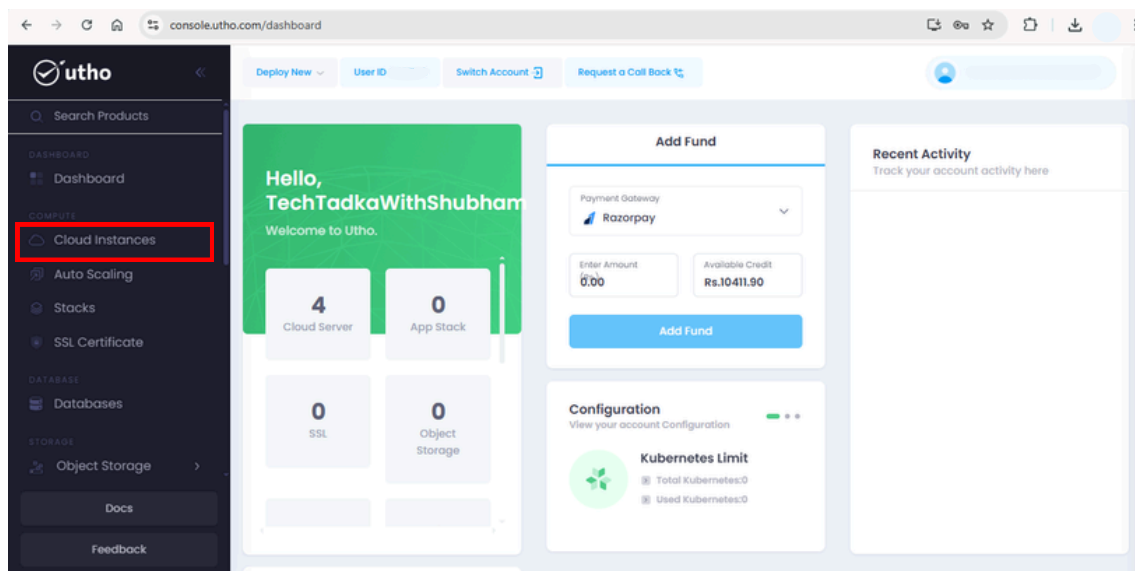
This project involves the deployment and configuration of the NGINX web server on Utho Cloud infrastructure. The project demonstrates fundamental cloud provisioning, system administration, and web server deployment practices.

The process included provisioning a Linux-based virtual machine, configuring network and firewall rules to allow SSH (Port 22) and HTTP (Port 80) access, installing and managing the NGINX service, and validating server accessibility via the public IP address.

## STEP-BY-STEP IMPLEMENTATION

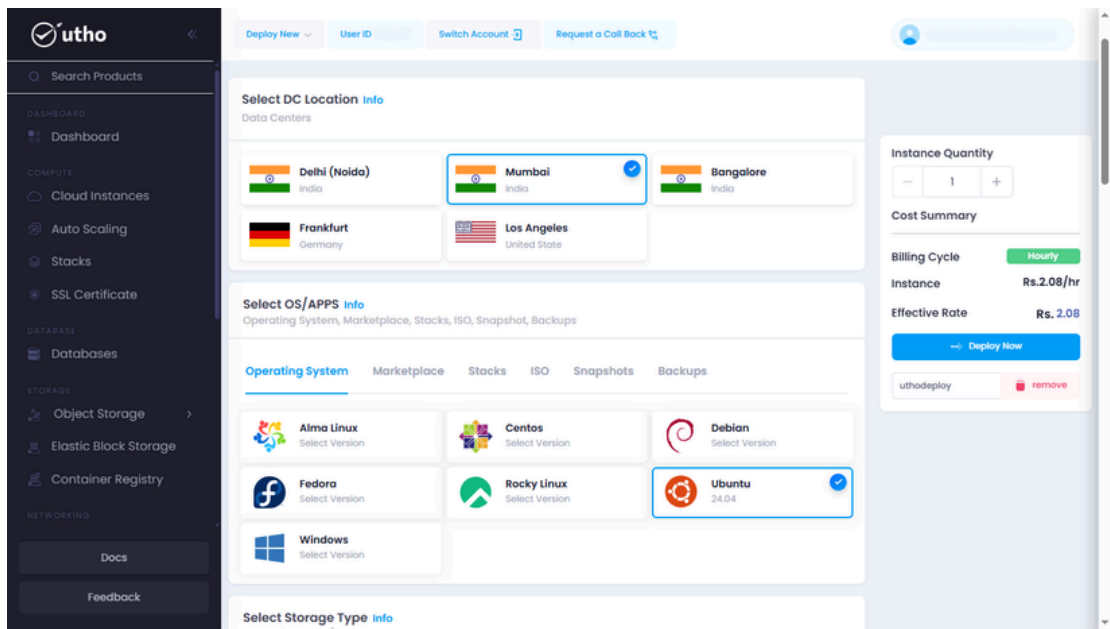
### Step 1: Launch a Cloud Instance

1. Log in to the Utho Cloud Console using your credentials.
2. Go to the Cloud Instances section from the dashboard and click Deploy New.

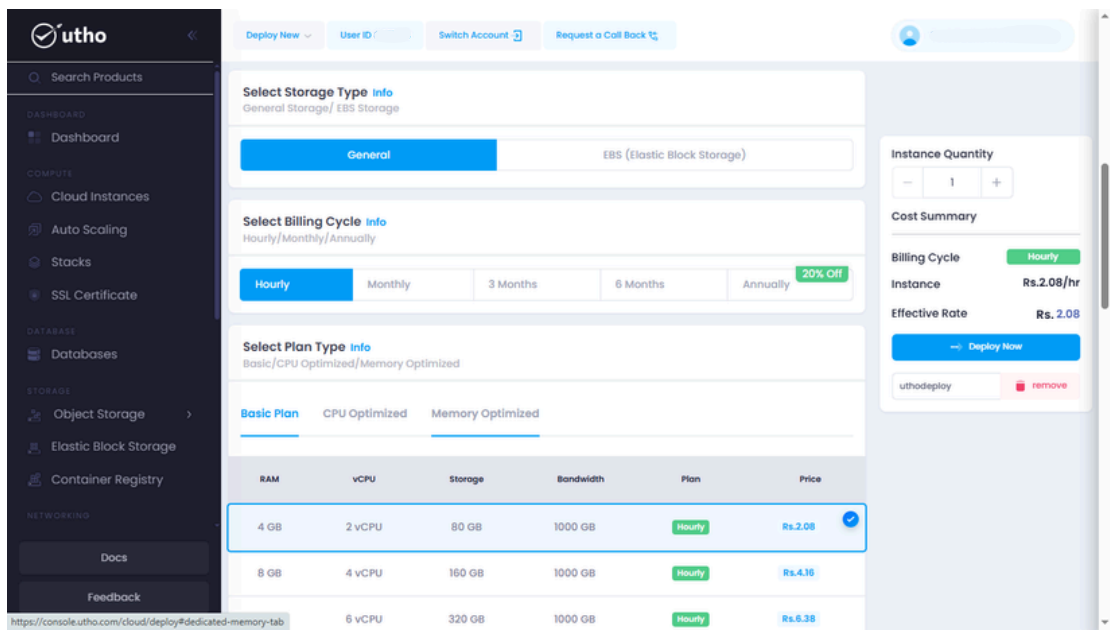


Step 2: Configure Cloud Settings:

- 1. **Select Datacenter Location:** Choose the DC Location for low latency. DC Location refers to the physical data center or region where cloud infrastructure is hosted.
- 2. **Select OS/APPS:** Choose your preferred Operating System (e.g. Ubuntu 24.04).



- 3. **Select Storage Type & Billing Cycle:** Choose the storage type and select the preferred billing cycle.
- 4. **Select Plan Type:** Choose the plan type and configure CPU, RAM, storage, and bandwidth as required.



5. **Setup Auth Configuration:** Setup the authentication method using either an SSH key or password.

6. **Deployment Type:** Select Deployment Type: Default (Without VPC) or With VPC.

The screenshot shows the utho dashboard with a sidebar on the left containing navigation links: Search Products, Dashboard, Cloud Instances, Auto Scaling, Stacks, SSL Certificate, Databases, Object Storage, Elastic Block Storage, Container Registry, Docs, and Feedback. The main content area is titled 'Auth Configuration Info' and 'Setup auth configuration'. It has two tabs: 'SSH Keys' (selected) and 'Password'. The 'SSH Keys' tab shows a text input field with a password icon and a 'Generate Pass...' button. The 'Password' tab shows a text input field and a 'Generate Pass...' button. Below this is the 'Deployment Type Info' section, titled 'Choose how you want to deploy your cloud instance -- with or without a VPC.' It has two options: 'Default (Without VPC)' (selected) and 'With VPC'. The 'Default (Without VPC)' option describes deploying with a public IP for quick setups. The 'With VPC' option describes deploying inside a VPC for private networking. At the bottom is the 'Security Group Info' section, titled 'Select a security group to control inbound and outbound traffic.', with a '+ Add New Security Group' button. On the right side, there is a 'Cost Summary' panel showing 'Instance Quantity' as 1, 'Billing Cycle' as 'Hourly', 'Instance' rate as 'Rs.2.08/hr', and 'Effective Rate' as 'Rs. 2.08'. It includes a 'Deploy Now' button and a 'uthodeploy' button with a 'remove' link.

7. **Select Security Group:** Select an existing group or create a new one to control inbound and outbound traffic

8. **Select CPU Model:** Select CPU model for the server.

9. **Hostname & Label:** Provide a hostname, review the configuration, and click Deploy Now button.

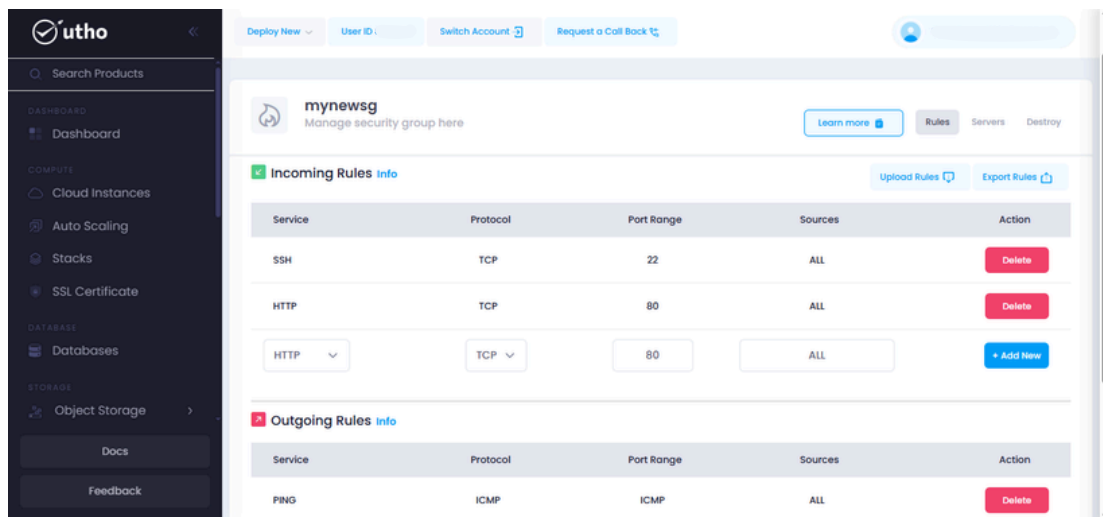
The screenshot shows the utho dashboard with the same sidebar as the previous image. The main content area is titled 'Security Group Info' and 'Select a security group to control inbound and outbound traffic.' It has a '+ Add New Security Group' button and a search bar 'Search by security group'. Below the search bar are five security group cards: 'basic-web-sg' (2026-02-27), 'my sg' (2026-02-27), 'coder sg' (2026-02-27), 'web sg' (2026-02-27), and 'mynewsg' (2026-02-27). Below this is the 'CPU Model Info' section, titled 'Select CPU model for the server', with two options: 'AMD' (selected) and 'intel'. At the bottom is the 'Hostname & Label' section, titled 'Name for the server instance', with a text input field containing 'My-Webserver'. On the right side, the 'Cost Summary' panel is the same as in the previous image, showing 'Instance Quantity' as 1, 'Billing Cycle' as 'Hourly', 'Instance' rate as 'Rs.2.08/hr', and 'Effective Rate' as 'Rs. 2.08'. It includes a 'Deploy Now' button and a 'uthodeploy' button with a 'remove' link.

10. Review all selected configuration details and confirm that all settings are correct. Click the **Deploy Now** button. The system will start provisioning the instance. This may take a few moments.

11. Once deployment is complete, the instance will appear on the Cloud Instances page. Note the Public IP assigned to the instance.

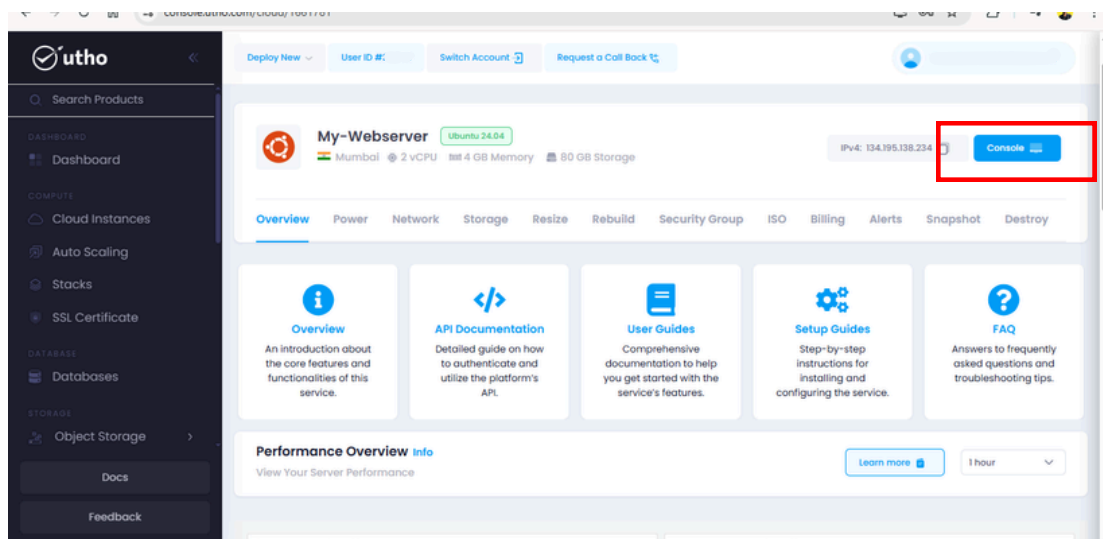
### Step 3: Configure Security Group Rules

1. Go to the Security Group section in the Utho Cloud dashboard and Click on the Manage button in front of your selected security group.
2. In Incoming Rules section, Click on the Add New button.
3. Add the following rules:
  - Port 22 (TCP): Allow SSH access
  - Port 80 (TCP) – Allow HTTP access
  - Set the source to ALL (0.0.0.0/0) to enable public access.



### Step 4: Access the Instance via Console Login

1. Go to your Instance and Click on console button to open the terminal.



2. When prompted, enter the username and password to log in to the server.

```
Ubuntu 24.04.3 LTS My-Webserver tty1
My-Webserver login: root
Password: _
```

3. After successful authentication, the terminal session will open, providing access to the Ubuntu server.

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@My-Webserver:~# _
```

- **To Connect to the Server via SSH**

1. Open the terminal on your local system.
2. Use the SSH command with public IP address to connect to the instance:
  - **ssh root@my-server-ip**
3. After authentication, a secure connection to the server will be established.

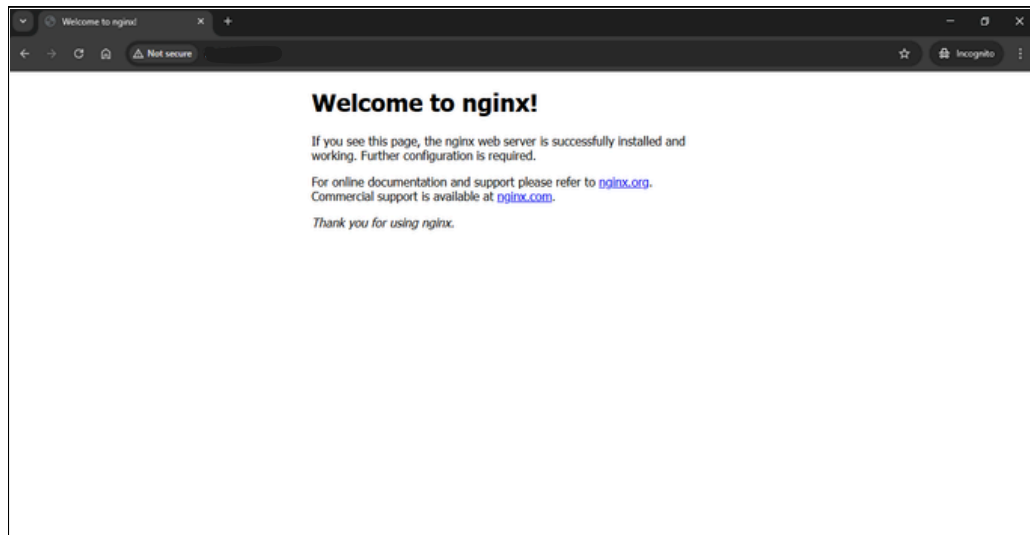
## **Step 5: Install and Verify NGINX**

1. After logging into the server, update the package repository to ensure the latest versions are available. Run the following command:
  - **sudo apt update**
2. Install the NGINX web server using the following command:
  - **sudo apt install nginx -y**
3. After installation, verify that the NGINX service is running:
  - **sudo systemctl status nginx**
4. Ensure the service status shows active (running).

## **Step 6: Verify NGINX Deployment in Browser**

1. Open a web browser on your local system.
2. Enter the public IP address of the instance in the address bar:

**http://<public-ip>**



3. If the installation is successful, the default NGINX Welcome Page will be displayed.

4. This confirms that the web server is successfully deployed and accessible over the internet.

## PROJECT OUTCOME

The deployment of the NGINX web server on Utho Cloud was successfully completed. This project demonstrated the complete workflow of provisioning a virtual machine, configuring security settings, installing a web server, and verifying public accessibility.

Through this implementation, a clear understanding of cloud instance management, firewall configuration, and basic Linux server administration was achieved. The successful display of the NGINX welcome page confirmed that the server was properly configured and accessible over the internet. This project serves as a foundational step toward more advanced cloud deployments and DevOps practices.